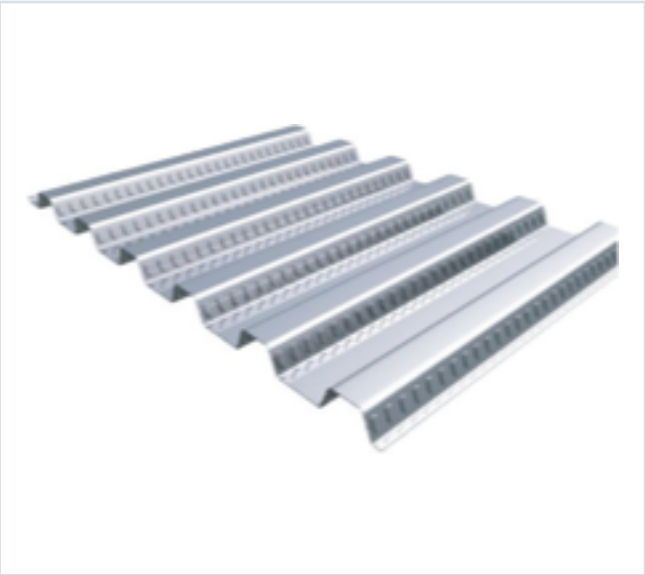


DECKING PRODUCT SUBMITTAL

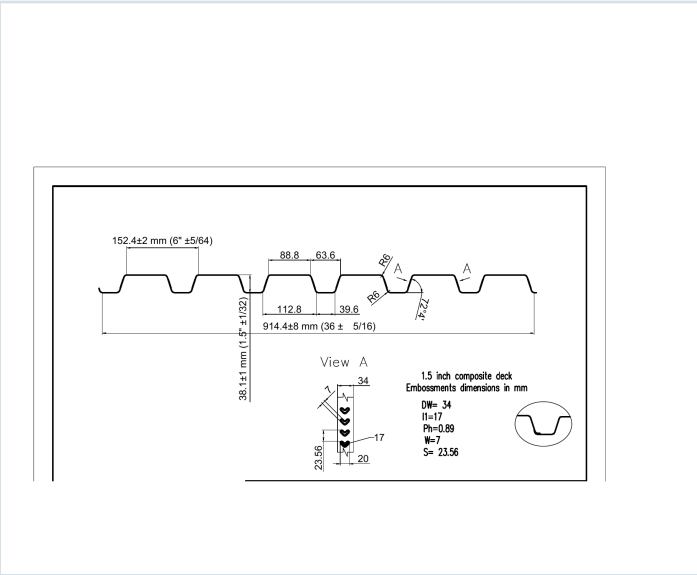
15CD-FY50-18-G40

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 50	18 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.6	50	0.294	0.310	8812	9291	0.257	0.290

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 45, 46, 47, 48
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

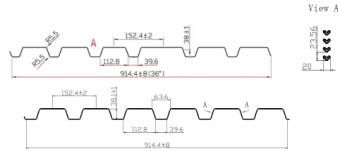
15CD-FY50-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 45

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (psf)	F _y (ksi)	S _x (in ³) per foot	S _y (in ³) per foot	ASD (φ = 1.67)		I _x (in ⁴) per ft.	I _y (in ⁴) per ft.
						M _u /Q (in ³ -lb/ft)	M _u /Q (in ³ -lb/ft)		
22	0.0295	16	50	0.170	0.077	583	538	0.144	0.107
20	0.0368	20	50	0.216	0.222	6617	6661	0.182	0.290
18	0.0474	26	50	0.294	0.330	8802	9208	0.257	0.290
16	0.0598	33	50	0.378	0.390	13327	14667	0.341	0.363

Note: All section properties and ASD/bearing strengths are calculated in accordance with AISI/SX 101-2017, AISI 508-2012 and AISI 508-2016.

Shear and Web Crippling

Gage	V _u /Q (lb/ft)	Web Crippling (R _u /Q) (lb/ft) One Flange Loading End Bearing	Web Crippling (R _u /Q) (lb/ft) Interior Bearing
22	2424	403	1015
20	3803	1643	1436
18	5052	194	2077
16	6279	205	3583

Note: All section properties and ASD/bearing strengths are calculated in accordance with AISI/SX 101-2017, AISI 508-2012 and AISI 508-2016.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	158	12	94	80	69	60	53	47	42	38	34
	20	272	142	100	102	88	77	67	60	55	50	45
	18	235	194	163	139	120	104	92	81	73	65	59
	16	202	250	207	179	154	134	119	105	93	82	73
Double	22	143	118	89	85	75	64	56	49	44	40	36
	20	198	147	125	105	91	79	69	61	55	49	44
	18	246	245	172	147	126	110	97	86	76	69	62
	16	311	257	216	184	159	138	122	106	96	86	78
Triple	22	179	148	124	106	91	79	70	62	55	49	45
	20	262	183	154	131	115	99	87	77	69	62	56
	18	310	256	215	183	158	138	121	107	96	86	77
	16	389	321	270	230	198	173	152	135	120	108	97

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Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	16	57	44	35	29	22	18	15	13	11	9
	20	16	72	56	44	36	28	23	19	16	14	12
	18	13	87	78	61	49	40	33	27	23	20	17
	16	179	135	104	82	65	53	44	36	31	26	22
Double	22	133	107	106	63	67	64	46	37	31	27	23
	20	230	173	133	105	84	68	56	47	40	34	29
	18	325	244	188	148	118	96	79	66	56	47	41
	16	431	324	250	196	157	128	105	88	74	63	54
Triple	22	143	107	63	65	52	42	35	29	25	21	18
	20	180	155	104	82	66	53	44	37	31	26	23
	18	254	191	147	116	95	75	62	52	44	37	32
	16	338	254	195	154	123	100	82	69	58	49	42

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

Construction Span Table - 20 psf Construction Load

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Total Slab Depth	Deck Type	Maximum Unstayed Clear Span	1 span	2 span	3 span	Total Slab Depth	Deck Type
3.0 (n=100) 31 PSF	15x16.22 ga	6' 0"	7' 0"	7' 9"		3.0 (n=100) 23 PSF	15x16.22 ga
	15x16.22 ga	7' 0"	8' 0"	8' 11"			15x16.22 ga
	15x16.22 ga	8' 0"	9' 0"	9' 11"			15x16.22 ga
	15x16.22 ga	9' 0"	10' 0"	10' 11"			15x16.22 ga
	15x16.22 ga	10' 0"	11' 0"	11' 11"			15x16.22 ga
4.00 (n=250) 43 PSF	15x16.22 ga	6' 0"	7' 0"	7' 9"		4.00 (n=250) 18 PSF	15x16.22 ga
	15x16.22 ga	7' 0"	8' 0"	8' 7"			15x16.22 ga
	15x16.22 ga	8' 0"	9' 0"	9' 7"			15x16.22 ga
	15x16.22 ga	9' 0"	10' 0"	10' 7"			15x16.22 ga
	15x16.22 ga	10' 0"	11' 0"	11' 7"			15x16.22 ga
4.50 (n=100) 43 PSF	15x16.22 ga	5' 0"	6' 0"	7' 0"		4.50 (n=100) 33 PSF	15x16.22 ga
	15x16.22 ga	6' 0"	7' 0"	8' 0"			15x16.22 ga
	15x16.22 ga	7' 0"	8' 0"	9' 0"			15x16.22 ga
	15x16.22 ga	8' 0"	9' 0"	10' 0"			15x16.22 ga
	15x16.22 ga	9' 0"	10' 0"	11' 0"			15x16.22 ga
5.00 (n=130) 49 PSF	15x16.22 ga	4' 0"	5' 0"	6' 0"		5.00 (n=130) 37 PSF	15x16.22 ga
	15x16.22 ga	5' 0"	6' 0"	7' 0"			15x16.22 ga
	15x16.22 ga	6' 0"	7' 0"	8' 0"			15x16.22 ga
	15x16.22 ga	7' 0"	8' 0"	9' 0"			15x16.22 ga
	15x16.22 ga	8' 0"	9' 0"	10' 0"			15x16.22 ga
5.50 (n=400) 55 PSF	15x16.22 ga	3' 0"	4' 0"	5' 0"		5.50 (n=400) 42 PSF	15x16.22 ga
	15x16.22 ga	4' 0"	5' 0"	6' 0"			15x16.22 ga
	15x16.22 ga	5' 0"	6' 0"	7' 0"			15x16.22 ga
	15x16.22 ga	6' 0"	7' 0"	8' 0"			15x16.22 ga
	15x16.22 ga	7' 0"	8' 0"	9' 0"			15x16.22 ga
6.00 (n=400) 61 PSF	15x16.22 ga	2' 0"	3' 0"	4' 0"		6.00 (n=400) 44 PSF	15x16.22 ga
	15x16.22 ga	3' 0"	4' 0"	5' 0"			15x16.22 ga
	15x16.22 ga	4' 0"	5' 0"	6' 0"			15x16.22 ga
	15x16.22 ga	5' 0"	6' 0"	7' 0"			15x16.22 ga
	15x16.22 ga	6' 0"	7' 0"	8' 0"			15x16.22 ga

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY50-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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LSC Deck - Wide Flange Up				ISNV Deck - Wide Flange Down			
50 ksi				50 ksi			
Live Load	20	psf	with concrete	Live Load	20	psf	with concrete
DL	160	psf		DL	160	psf	
Deck Weight	2	psf		Deck Weight	2	psf	
Live Load	50	psf	without concrete	Live Load	50	psf	without concrete
Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC	Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC
3.5	22	36	36	3.5	22	25	25
3.5	20	31	31	3.5	20	30	30
3.5	18	40	40	3.5	18	38	38
3.5	16	47	47	3.5	16	46	46
4	22	36	36	4	22	25	25
4	20	31	31	4	20	30	30
4	18	40	40	4	18	38	38
4	16	47	47	4	16	46	46
4.5	22	36	36	4.5	22	25	25
4.5	20	31	31	4.5	20	30	30
4.5	18	40	40	4.5	18	38	38
4.5	16	47	47	4.5	16	46	46
5	22	36	36	5	22	25	25
5	20	31	31	5	20	30	30
5	18	40	40	5	18	38	38
5	16	47	47	5	16	46	46
5.5	22	36	36	5.5	22	25	25
5.5	20	31	31	5.5	20	30	30
5.5	18	40	40	5.5	18	38	38
5.5	16	47	47	5.5	16	46	46
6	22	36	36	6	22	25	25
6	20	31	31	6	20	30	30
6	18	40	40	6	18	38	38
6	16	47	47	6	16	46	46

Assumes that backspan is greater than the cantilever spans.
Applicable to single, double, triple spans or greater.
Controlled by concentrated load = 30 psf
Controlled by concrete weight = 30 psf
Controlled by Deflection

STEEL CATALOG PAGE 48

22 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	372	334	289	252	210	170
4	37	400	400	400	397	340	298	258	218
4.5	43	400	400	400	400	400	397	331	281
5	49	400	400	400	400	400	400	369	319
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	277	156	135	121	110	96	85	80
4	224	197	175	156	139	125	113	102
4.5	273	241	213	190	170	153	138	126
5	323	286	253	226	200	182	164	149
5.5	375	331	294	262	235	211	191	173
6	400	378	336	300	269	242	218	197

20/18/16 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	400	390	337	285	246	207
4	37	400	400	400	400	400	398	332	282
4.5	43	400	400	400	400	400	400	381	331
5	49	400	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	236	181	170	151	136	122	110	100
4	274	242	215	192	172	155	140	127
4.5	335	296	263	235	215	190	172	156
5	398	362	315	280	251	226	203	186
5.5	400	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Note
Due to the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gage. However, the construction square do get longer for 18 and 16 gage deck.